

Renewable Sources of Energy

3.1 ENERGY MANAGEMENT AND CONSERVATION

3.1.1 Energy management

Definition

Energy management is planning and operation of energy production and energy consumption units as well as energy distribution and storage.

For the past decades, energy generation has been shifted to alternative energy sources like renewable energy forms such as solar, wind and biomass energy, etc., instead of the conventional fossil fuel sources. Apart from the growth in the energy sector, there has been an equivalent increase in business and organizations, which has brought tremendous competition in the market in terms of increasing environmental standards and reducing global warming, carbon foot print and green house gas emissions.

Energy management introduced in an organization, can effectively manage how much energy they produce and how to controls and monitor. A large amount of energy and money can be saved by employing energy management principles. It can also help companies by not only improving productivity but also the quality that they offer using energy efficiency techniques and better materials and manufacturing processes.

Objectives (or) Aims of energy management

- (i) Resource conservation.
- (ii) Climate protection.
- (iii) Cost savings / minimize waste.
- (iv) Minimize environmental effects.

The ultimate aim (or) objectives of this process is not only to save the cost but also to achieve complete environmental sustainability.

Principles of energy management

1. It controls the costs of the energy function and not Btu of energy.
2. The second principle is to control energy functions as a product cost.
3. The third principle is to control and meter only the main functions, which accounts for only 20% functions which make up 80% of the costs.
4. The last principle states that the major effort of an energy management program should be put into installing controls and achieving results.

Steps involved in the process of energy management

The following 5 steps are important in the process of energy management.

- Step 1:** Collecting and analyzing continuous data.
- Step 2:** Identify optimizations in equipment schedules, set points and flow rates to improve energy efficiency.
- Step 3:** Calculate return on investment. Units of energy saved can be metered and calculated just like units of energy delivered.
- Step 4:** Execute energy optimization solutions.

Step 5:

Repeat step 2 to continue optimizing energy efficiency.

3.1.2 Energy conservation

Definition

Energy conservation is the practice of using less energy in order to lower the costs and reduce environmental impact.

This can be achieved either by using

- (i) energy more efficiently (using less energy for a constant service),
- (ii) by reducing the amount of service used (by driving less).

Objectives (or) Aims of energy conservation

Energy conservation is the key element. The main objectives are

- (i) to reduce overall energy demand,
- (ii) to lower energy cost,
- (iii) to reduce energy consumption,
- (iv) to lower the overall green house gas emission.

Principle (or) Law of conservation of energy

The principle of energy of conservation states that energy can neither be created nor destroyed but it can be transformed from one type to another.

According to this the total energy of an isolated system remains constant.

Importance of energy conservation

Energy conservation is very important because of the following reasons.

1. It reduces our usage of non-renewable energy resources (like fossil fuels).
2. It also helps you to save money on energy costs including utility bills and other energy bills.
3. It also cuts down on expanding development, where natural resource extraction is impacting natural areas.
4. When we conserve energy more efficiently, we directly reduce the amount of green house gas emissions entering the earth's atmosphere.
5. It insists us to replace the energy, used, with an alternate energy source.

15 ways to conserve energy (or) conservation

There are 15 ways to start conserving energy.

1. Adjust your day-to-day behaviors

Example

Switch off lights (or) appliances when you do not need them.

2. Replace your light bulbs

Traditional incandescent light bulbs consume more amount of electricity and must be replaced by energy efficient alternatives like CFL and LED bulbs.

3. Use smart power strips

"Phantom loads" (or) the electricity used by electronics, when they are turned off (or) stand by mode, are the major source of energy waste. Smart power strips, eliminate this problem, when they are not in use.

4. Install a programmable (or) smart thermostat

It automatically turn-off (or) reduce heating and cooling during the time when you are asleep (or) away.

6. Purchase energy efficient appliances

These will consume less energy during use.

7. Reduce your water heating expenses

Efficient water heaters can be 8% to 300% more energy efficient than a conventional storage water heater.

8. Install energy efficient windows

To prevent heat loss through your windows,

(i) you can replace single-pane windows with double-pane windows,

(ii) gas filled windows with "low-e" coatings can be fixed.

9. Upgrade your HVAC system

Most energy efficient way to upgrade your home's HVAC system is installing air source heat pump. In order to heat and cool your home a heat pump transfers heat from one place to another.

10. Weatherize your home

Air leaks into your home are windows, doors and vents. To prevent these leaks, care must be taken in such a way that no cracks (or) opening between the walls and windows is present.

11. Insulate your home

Insulation retains heat during the winter and keeping heat out of your home during the summer.

11. Wash your clothes in cold water.

12. Replacing dirty air filters regularly can reduce energy consumption upto 15%.

13. As microwave is more energy efficient, microwave oven can be used instead of ordinary stove.

14. Using natural light, like sun, we can reduce the energy consumption.
15. Dress appropriately for the weather inside and outside.

3.2 NEW ENERGY SOURCES

3.2.1 Need

Fossil fuels and nuclear energy are the important resources used to meet most of our energy needs today. These are expected to be widely used in the near future. However, fossil and nuclear energy resources are non-renewable and will someday be exhausted, while their continued use possess environmental risks related to air pollution, global climate change, land use and waste disposal. These issues have stimulated the search for new energy sources for producing and using energy.

3.2.2 Different types of new energy sources

New energy sources (or) Renewable energy resources that are being developed include

- (i) Hydrogen energy.
- (ii) Ocean thermal energy conversion.
- (iii) Tidal and wave energy.
- (iv) Geothermal energy.
- (v) Solar energy.
- (vi) Wind energy.
- (vii) Bio-mass energy.
- (viii) Artificial Intelligence (AI).
- (ix) Photovoltaics (PV).
- (x) Distributed energy storage systems (DESS).

- (xi) Grid integration.
- (xii) Space technologies.
- (xiii) Norwegian Crystals.
- (xiv) Algal bio-fuels.
- (xv) Body heat.
- (xvi) Dance flowers.

1. Hydrogen

The fuel that has potential of being widely used in the future is hydrogen gas (H_2). Like natural gas hydrogen can be burned to heat buildings, cook food and produce electricity in power plants.

Hydrogen possess high calorific value. It is non-polluting, because the combustion product is water.



H_2 gas can be compressed in a fuel tank and used to power cars and buses.

Sources of hydrogen

- (i) Plentiful hydrogen is available from water (H_2O). Water can be split into gaseous H_2 and O_2 by an electrolysis process.
- (ii) Hydrogen can also be produced from natural gas and biomass resources.
- (iii) Ethanol reacts with high-temperature steam to produce hydrogen.
- (iv) Biomass is converted into sugar-rich feed stocks that can be fermented to produce hydrogen.
- (v) Microbes such as green algae, consume water in the presence of sun light and produce hydrogen as a by-product.

Hydrogen fuel cell

Hydrogen can be used in fuel cells. The electrons in hydrogen atoms generate electricity in the fuel cell. The combination of H_2 and O_2 creates water and heat from the reaction. The heat may be used to produce electricity.

At anode, hydrogen is split into protons and electrons. The electrons moves to cathode and generates electricity.

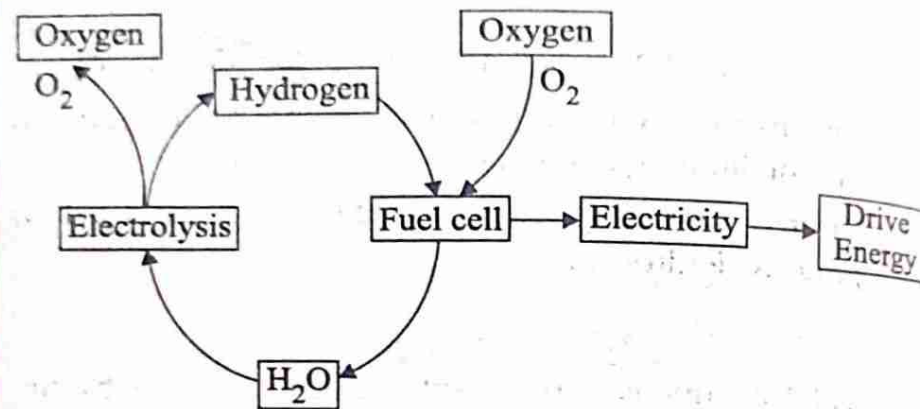


Fig. 3.1 Hydrogen fuel cell Technology

Electrical power plants can be built using large banks of fuel cells, but small groups of cells provide electricity for individual home and commercial buildings.

Problem

- (i) Difficulties in storing enough hydrogen for motor vehicles to run long distances.
- (ii) Infrastructure to refuel these vehicles.
- (iii) Highly inflammable and explosive in nature.
- (iv) Safe handling is required.

2. Ocean thermal energy (OTE)

There is often large temperature difference between the surface level and deeper level of the tropical oceans.

This temperature difference can be utilized to generate electricity. The energy available due to the difference in temperature of water is called ocean thermal energy.

Condition

The temperature difference should be of 20°C (or) more is required between surface water and deeper water.

Significance of OTE

- (a) OTE is continuous, renewable and pollution free.
- (b) The use of cold deep water, as the chiller fluid in air-conditioning, has also been proposed.
- (c) Electric power generated by OTE can be used to produce hydrogen.

3. Tidal energy (or) Tital power

Tidal energy is a renewable energy powered by the natural rise and fall of ocean tides and currents.

Significance of tidal energy

- (i) Tidal power plants do not require large areas of valuable lands as they are on the bays (or) estuaries.
- (ii) As the sea water is inexhaustible, it is completely independent of the uncertainty of precipitation (rainfall).
- (iii) It is pollution-free energy source, as it does not use any fuel and also does not produce any wastes.

4. Geo-thermal energy

The heat produced deep in the Earth's core is called Geothermal energy. The energy harnessed from the high temperature present inside the earth can be used to produce electricity.

Significance of geothermal energy

- (i) The power generation level is higher for geothermal than for solar and wind energies.
- (ii) Geothermal power plants can be brought on line more quickly than most other energy sources.
- (iii) GTE is effectively and efficiently used for direct uses such as hot water bath, resorts, aquaculture, greenhouses.

5. Solar energy

Solar energy is derived by capturing radiant energy from sunlight and converting it into heat, electricity (or) hot water.

Significance of solar energy

- (i) Solar cells are noise and pollution free.
- (ii) Solar water heaters, cookers, require neither fuel nor attention while cooking food.
- (iii) Solar cells can be used in remote and isolated areas, forests, hilly regions.

6. Wind energy

Moving air is called wind. Energy recovered from the force of wind is called wind energy. The energy possessed by wind is because of its high speed. The wind energy is harnessed by making use of wind mills.

Significance of wind energy

- (i) The generation period of wind energy is low and power generation starts from comissioning.
- (ii) It is recommended to broaden the nation's energy options for new energy sources.
- (iii) It is made available easily in many off-shore, on-shore and remote areas.

Bio-mass energy

Biomass is the organic matter, produced by plants (or) animals, used as sources of energy. Most of the biomass is burned directly for heating, cooling and industrial purposes.

Examples Wood, crop residues, seeds, cattle dung, sewage, agricultural wastes, etc.,

Significance of bio-mass energy

1. The cost of obtaining bio-energy through bio-gas plant is less than the cost of obtaining energy from fossil fuels.
2. Biomass consumes more CO_2 than is released during combustion of biomass.
3. It provides a stored form of energy and in many cases in a form suitable for vehicle propulsion.

8. Artificial intelligence (AI) in the energy sector

Artificial intelligence (AI) is used to forecast demand and manage the distribution of resources, to ensure that power is available at the time and place it's needed with a minimum of waste. AI plays an essential role in the world's transition to clean energy. Artificial intelligence is particularly important in the renewable energy industry, where it often can't be stored for long periods of time and has to be used close to the time and location where it is generated.

9. Photo Voltaics (PV)

Solar companies are integrating PV systems to minimize the need for additional land usage. As a result, integrated PV, floatovoltaics and agrivoltaics are logical shift in trends. Now thin film PV cells are being developed

to make solar panels flexible, cost-effective, light weight and environment friendly.

10. Distributed energy storage systems (DESS)

A distributed energy storage system (DESS) is a packaged solution that stores energy for use at a later time. The system is provided with two main components,

- (i) DC charged batteries and
- (ii) Bi-directional inverter.

It's major role is to prevent power fluctuation and power quality problems.

11. Grid integration

Grid integration is the practice of developing efficient ways to deliver variable renewable energy to the grid.

12. Space technologies

Space-based energy technologies like

- (i) harvesting hydrogen from the moon to power fuel cell on earth,
- (ii) orbiting solar rays that absorb around-the-clock direct sunlight and send the energy back down to stations on the ground via radio (or) microwaves,

are being developed as the new energy technology.

13. Norwegian crystals

- (i) Low carbon mono crystalline silicon ingots, is a type of crystal, used for high performance photovoltaic devices.
- (ii) Gallium-doped ingots, that increases the lifetime of the solar cells.

Through this, Norwegian crystals controls the carbon footprint of solar panel components at ultra low levels.

14. Algal bio-fuels

Algae is an alternative to liquid fossil fuels. It offers huge commercial potential. Like fossil fuel, algae fuel releases CO_2 when burnt. But unlike fossil fuel, the CO_2 released by algae fuel is removed from the atmosphere via photosynthesis as the algae (or) plant grew.

The impact of algal bio-fuels on the atmosphere is much lower. Algal fuel production has a minimal impact on land and water resources. It can be produced using seawater (or) even grey waste water.

15. Body heat

The body heat, liberated by humans in the crowded area like central station, market place, is channelled through the station's vent system. Then it is used to warm up water in underground tanks and pumped through the heating system.

16. Dance floors

The kinetic energy of the dance floor is converted to electricity that lights up the dance floor itself.

Pavegen, a london based company is demonstrating this with its development of the energy harvesting "smart street".

3.3 APPLICATIONS OF HYDROGEN ENERGY

1. Hydrogen is a reagent, used in many industries, including chemicals, textile fiber manufacturing, glass, electronics and metallurgy.
2. It is also used as a fuel for rocket launchers.

3. In electronics, hydrogen is used as a carrier gas, for the manufacture of electronic components.
4. Hydrogen is used in industries for many applications.

Examples

- (i) It combines with nitrogen to produce ammonia, a base for fertilizers.
 - (ii) It is a good reagent for textile fibers like nylon, polyurethane foam.
5. Hydrogen is used in metallurgy for heat treatment process to produce mechanical parts (or) to alter their properties.
 6. Hydrogen is used during fuel refining to remove this sulphur via a process of desulphurization.
 7. Atomic hydrogen welding (AHW) is a type of arc welding which utilizes a hydrogen environment.
 8. A mixture of hydrogen and nitrogen is used to prevent oxidation in flat glass production.
 9. As an efficient reducing and etching agent, hydrogen is used to create semiconductors, LEDs, displays.
 10. Hydrogen gas is used as a therapeutic gas for a number of different diseases.

11. Hydrogen fuel cell

Hydrogen fuel cell uses hydrogen as a fuel in an electrochemical process that combines H_2 and O_2 to produce electrical energy with water and heat as the only by-product. Two main applications of fuel cells are,

(i) Stationary power sources.

- (a) These are used to power office buildings, data centres, grocery stores and off-grid telecommunication towers.

- (b) It is used as a part of uninterruptible power supply (UPS) system, where continuous uptime is critical.

(ii) Hydrogen fuel cell vehicles (FCVs)

- (a) The heat produced by the hydrogen fuel cell can be used for space and water heating (or) industrial process.
- (b) Hydrogen fuel cells power clean trucks, fork lifts, etc.
- (c) Hydrogen power is being considered for transportation applications including hydrogen fuel cell buses.
- (d) Hydrogen fuel cell trains have now appeared.
- (e) Hydrogen offers versatile options for mobile power generation. Some of the hydrogen fuel cells were developed by NASA to provide electricity for rockets and shuttles in space.
- (f) Hydrogen fuel cells have found a number of marine applications (used in boats and submarines).

3.3.1 Advantages and disadvantages of hydrogen fuel cells

Advantages

1. Hydrogen is readily available.
2. It does not produce harmful emissions.
3. It is environmentally friendly.
4. It can be used as fuel in rocket.
5. It is energy efficient and more powerful than fossil fuels.
6. It is renewable.
7. It reduces carbon foot prints.

8. Charging times is fast.
9. It does not make noise and visual pollution.
10. It can be used for long time.

Disadvantages

1. It is expensive.
2. It is difficult to store.
3. It is highly inflammable.
4. Infra-structure.
5. Regulatory issues.

3.4 APPLICATIONS OF OCEAN ENERGY RESOURCES

Tidal energy (or) tidal power is a form of ocean energy that is harnessed by converting tide energy into useful forms of power. Tidal energy is obtained from the rise and fall of tides. Tidal barrages and dams are constructed across a narrow opening to the sea. Water rushes into the dam when the sea level rises which moves the blade of the turbines which helps in the generation of electricity.

Below are some important applications of ocean energy.

1. Ocean waves

Potential energy associated with ocean waves can be harnessed using modular technologies.

2. Temperature gradients

Difference in thermal energy between sea surface and deep water can be harnessed by Ocean Thermal Energy Conversion (OTEC) process.

3. Salinity gradient

At the mouth of rivers, where fresh water mixes with salt water, energy associated with the salinity gradient can be harnessed using pressure retarded reverse osmosis process and associated conversion technologies.

4. Ocean wave energy convertors

These are the technology used to trap the mechanical energy of the wave to convert it to electrical power.

5. Oscillating bodies

Oscillating waves uses hydraulic motors (or) electrical generators as a power take-off system.

6. Overtopping wave energy convertors

The overtopping wave energy convertors (or) terminators, using the low head hydraulic turbines, converts the potential energy formed by the height of accumulated water over the wave surface to electrical power.

7. Ocean thermal energy (OTE)

The temperature difference, between the surface level and deeper level of the tropical oceans, can be utilized to generate electricity.

3.4.1 Advantages and disadvantages of ocean energy

Advantages (or) Benefits

- (i) Ocean energy is cheaper and efficient.
- (ii) It is environment - friendly.
- (iii) The source of ocean energy is inexhaustible
- (iv) Operational and maintenance costs are low.
- (v) Tidal energy sources can last for decades.

- (vi) It protects coastal floodings due to the stability of rock armor.

Disadvantages (or) Limitations

- (i) Construction of tidal power plant is expensive and requires high capital investment.
- (ii) Maintenance and equipment repairing is a challenge.
- (iii) Negative influence on marine life forms.
- (iv) Storage capacity is required.
- (v) Environmental problems like habitat change arises.

3.5 APPLICATIONS OF TIDAL ENERGY CONVERSION

1. Electricity can be generated from the tidal energy.
2. Tidal energy is used in grinding mills for the mechanical crushing of grains.
3. Tidal energy is used to rotate a turbine.
4. Tidal energy is used to store energy in a hydroelectric dam, acting as large energy storage.
5. Tidal barrages and reservoirs can be modified to store energy.
6. Tidal barrages are capable of preventing damages to the coast during high storms.
7. Tidal barrages also help to create easy transport between the two arms of an estuary (or) a bay.

3.5.1 Advantages and disadvantages

Advantages (or) Merits

1. It is environment friendly.
2. It is cheaper and efficient.
3. Low operating and maintenance costs.

4. The source of energy is inexhaustible.
5. Protects coastal flooding.
6. Tidal energy sources can last for decades.
7. Power output is highly predictable.

Disadvantages

1. Construction of tidal power plants is expensive and requires high capital investment.
2. Equipment repairing and maintenance is difficult.
3. Environment problems, like habitat change, arises.
4. Storage capacity is required.
5. Negative influence on marine life forms.
6. Location limited.

3.6 GEOTHERMAL POWER PLANT (GTE)

3.6.1 Definitions

1. Geothermal Power

It is the electrical power generated from geothermal energy.

2. Geothermal Energy

It is the heat produced deep in the earth's core.

3.6.2 Origin

Geothermal energy is the thermal energy found in the earth's crust which originates from the formation of the planet and from radioactive decay of materials. The high temperature and pressure in earth's interior cause some rock to melt and solid mantle to behave plasmically. This results in parts of the mantle convecting upward since

it is lighter than the surrounding rock. Temperatures at the core mantle boundary can reach over 400°C.

3.6.3 Concept

Geothermal technology extracts the heat found within the subsurface of the earth, which can be used directly for heating and cooling (or) converting it to electricity.

The steam comes from the reservoirs of hot water, found a few miles (or) more below the earth's surface, rotates a turbine that activates a generator, which produces electricity.

3.6.4 Power plants of GTE

Geothermal power plant uses hydrothermal resources that have both water (hydro) and heat (thermal). Geothermal power plants requires high temperature (300°F to 700°F) hydrothermal resources that come from either dry steam wells (or) from hot water wells.

Generally we use these resources by drilling wells into the earth and then piping steam (or) hot water to the surface. The hot water (or) steam rotates a turbine that generates electricity. The depth of the geothermal wells is as much as 2 miles.

Types of geothermal power plants

There are three basic types of geothermal power plants.

1. Dry steam power plant

It uses steam directly from a geothermal reservoir to drive generator's turbines.

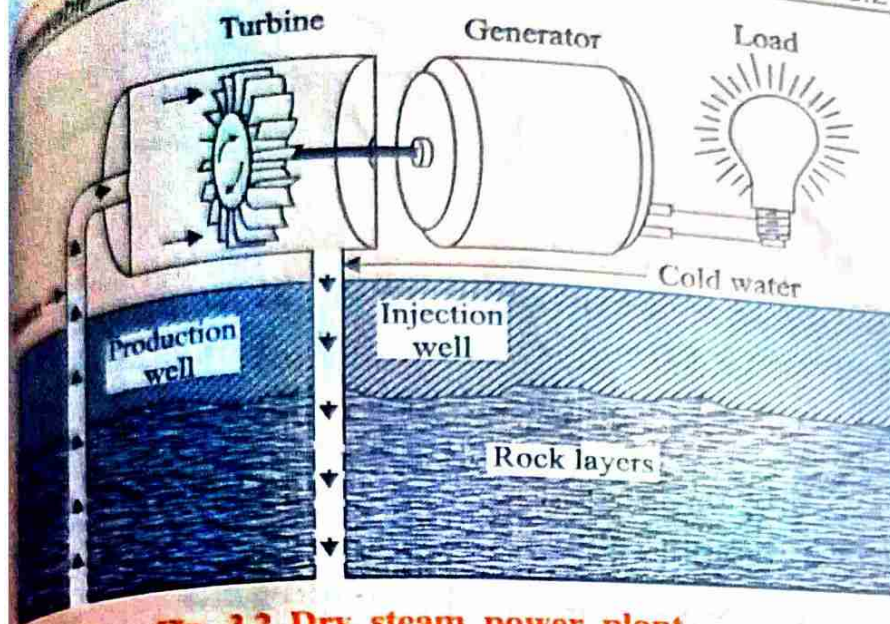


Fig. 3.2 Dry steam power plant

Flash steam power plant

It takes high-pressure hot water from deep inside the earth and converts it into steam to drive generator's turbine.

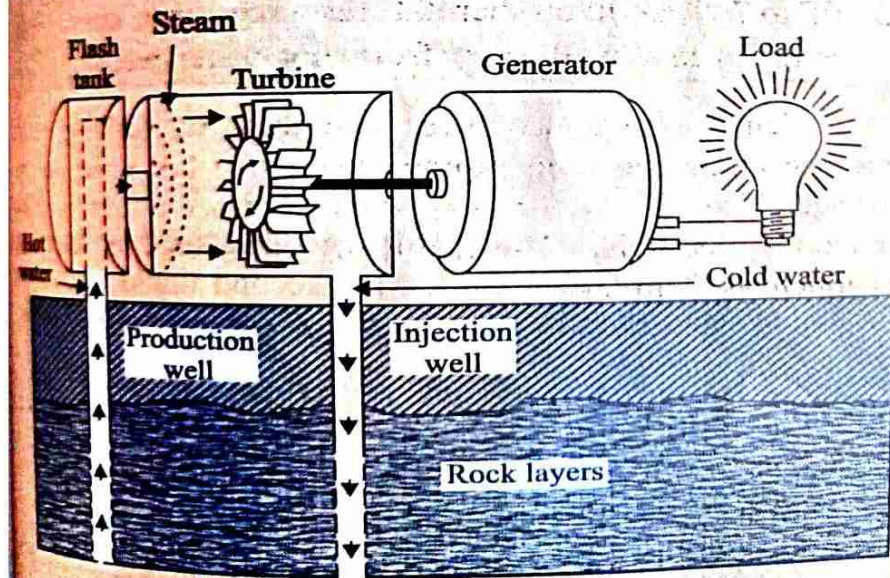


Fig. 3.3 Flash steam power plant

When the steam cools, it condenses to water and is injected back into the ground to be used again. Most geothermal power plants are flash steam plants.

3. Binary cycle power plants

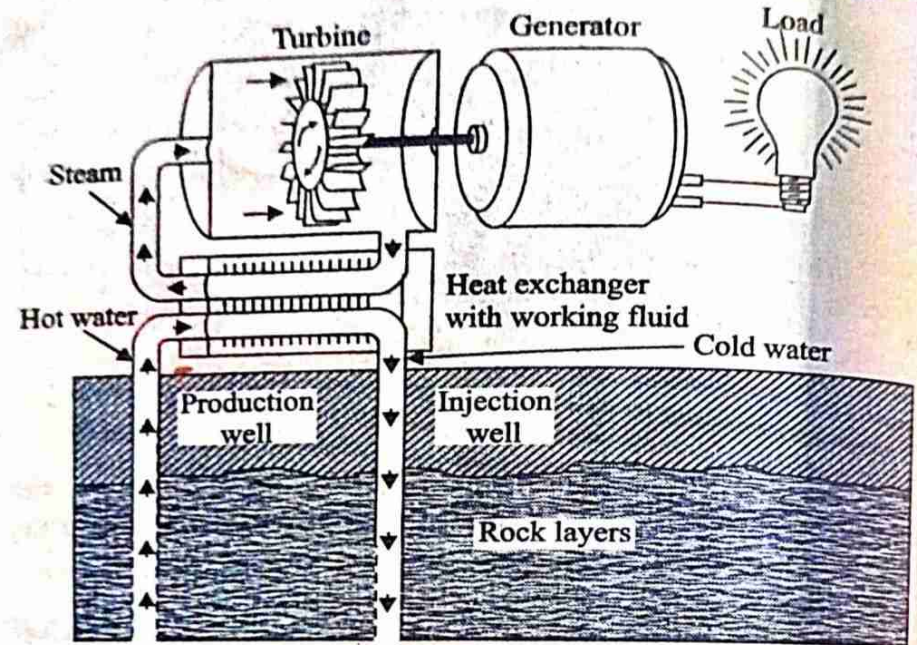


Fig. 3.4 Binary cycle power plants

It transfers the heat from geothermal hot water to another liquid. The heat causes the second liquid to convert it into steam, which is used to drive a generator's turbine.

3.6.5 Advantages and disadvantages of GTE

Advantages

1. GTE is environmentally friendly.
2. GTE is a source of renewable energy.
3. It is the sustainable form of energy.

4. The potential of GTE is huge.
5. Energy generated from this resource is reliable.
6. As GTE is natural, no fuel is required.

Disadvantages

1. Location is restricted.
2. As GTE does not release green house gases, there are many other gases released into the atmosphere (cause side effects).
3. May cause earthquakes.
4. It is expensive resource.
5. Management is required to maintain sustainability.

3.6.6 Applications of GTE

1. GTE is used for space heating and cooling.
2. GTE is used to generate electricity.
3. It is also used for industrial process heat.
4. It is used for desalination of geothermal water and heavy water production.
5. It is also used in the extraction of minerals from geothermal fluids.
6. Geothermal Heat Pumps (GHPs) are used to heat buildings in the winter and cool them in summer.
7. The direct use of GTE involves the use of heated water from the ground without the need for any other sources.